GROUP THEORY

Scanned by CamScanner

Group :⇒

Def^n : Algebraic Structure :

A set of an objects or elements with mathematical operation such as addition, multiplication, union and intersection etc. associated with it's properties called as algebraic structure.

Let S be the set and $*$ be the mathematical operation then $(S, *)$ represent an algebraic structure

Binary operation :

Let $*$ an operation $*$ define on set S. is a function mapping from $S \times S$ to S

ie $* : S \times S \longrightarrow S$

Such that $\forall\ a, b \in S$, $*(a,b) = c \in S$ (codomain set.)

ie $\boxed{a * b = c \in S}$

then $*$ is called as binary operation.

Note: algebraic operations, addition and multiplication is are binary operation's. But subtraction & division are not binary operations.

For eg ① $\forall\ a, b \in \mathbb{N}$ since $a+b \in \mathbb{N}$
(∵ $+$ is binary operation)

② $\forall\ a, b \in \mathbb{N}$ since $a \cdot b \in \mathbb{N}$ ∵(• multiplication is binary operation

But $\forall\ a, b \in \mathbb{N}$ since $a - b \notin \mathbb{N}$ ~~as well as~~
(if $a<b$)
(∵ '−' is not binary operation.)

Scanned by CamScanner

Some important properties

① **Closure property.**

An operation $*$ define on set A is said to be satisfying closure property.

if for all $a, b \in A, \Rightarrow (a*b) \in A$

and set A is said to be closed w.r.to. operation $*$

for ex ① let R be the set of Real number

and since for $x, y \in R$

$\Rightarrow x+y \in R$

$\therefore$ operation '+' satisfy closure property

and set R is closed w.r.to addition.

---

② **Associative property:**

A binary operation '$*$' define on set A. is said to be satisfying Associative property

if for every $a, b, c \in A$

$$a*(b*c) = (a*b)*c$$

Illustration:

① Addition and multiplication of real number are associative always

ie $\forall\ a, b, c \in R$

(i) $a+(b+c) = (a+b)+c$

(ii) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$

Scanned by CamScanner

But the operation subtraction (−) define on set of Integers is not an associative

because for any $a, b, c \in I$

$$a-(b-c) \neq (a-b)-c$$

Similarly Operation division (÷) define on set of integers is also not an associative.

because for any $a, b, c \in I$

$$a \div (b \div c) \neq (a \div b) \div c$$

Existance of Identity element

Let $e \in A$ is an element of set A

then e is said to be an identity element of set A

if $\forall\ a \in A$, $\boxed{a * e = a}$ or $\boxed{e * a = a}$

Illustration:

let N be the set of Natural number

ie $N = \{1, 2, 3, 4, 5 \ldots\ldots\}$

and since $\forall\ a \in N$, $a \cdot 1 = a = 1 \cdot a$

and also $1 \in N$

$\Rightarrow$ 1 is identity element of N under the operation multiplication

$\therefore$ 1 is multiplicative identity of N.

Now. $\forall\ a \in N$. Since $a + 0 = a = 0 + a$

but $0 \notin N$.

$\Rightarrow$ 0 is not an identity element of N under the operation addition '+'

Scanned by CamScanner

In addition to that

If operation '+' define on set of ~~~~ Real number then $0 \in R$ is said to be identity of set R. because $\forall\ a \in R$

$$\Rightarrow a + 0 = a = 0 + a$$

Hence $0 \in R$ is called additive identity of R.

---

(4) ~~Exa~~ Existance of Inverse:

let $e \in A$ is and identity of ~~element~~ set A under the operation '*'

and ~~if~~ for any $a, b \in A$

if we have $\boxed{a * b = e = b * a}$ where $e \in A$

Then a is inverse of b and vice versa.

Illustration:

Let R be the set of real number and we know that $0 \in R$ is additive identity of set R under the operation '+'.

also if $a \in R$ then $(-a) \in R$

and $a + (-a) = 0 = (-a) + a$

$\Rightarrow -a$ is inverse of a with respect to operation '+'

and vice versa.

Scanned by CamScanner

Similarly: let R be the set of real number.

then we know that $1 \in R$ is multiplicative identity (1 is identity of set R w.r.t. operation '$\cdot$')

also for any $a \in R \Rightarrow \frac{1}{a} \in R$

and since $(a) \cdot \left(\frac{1}{a}\right) = 1 = \left(\frac{1}{a}\right) \cdot (a)$

$\Rightarrow \frac{1}{a}$ is inverse of a w.r.t. to multiplication operation.

and vice versa.

Note: Inverse of Identity element is Identity itself.

(5) **Commutative property:**

A binary operation '$*$' ~~defined~~ define on set A is said to be commutative

if $\forall\ a, b \in R$, $\boxed{a * b = b * a}$

Illustration:

① addition and multiplication of real numbers are commutative always

ie $\forall\ a, b \in R$, $a + b = b + a$

& $a \cdot b = b \cdot a$

But subtraction and division of real number are not commutative always.

$\because \forall\ a, b \in R$, $a - b \neq b - a$

& $a \div b \neq b \div a$

Scanned by CamScanner

⑤ Distributive law :

If '*' and ⊕ are two binary operations define on set A. and it satisfying the following condn. such as

$$a * (b \oplus c) = (a * b) \oplus (a * c)$$ [Left Distribution]

Similarly $$(b \oplus c) * a = (b * a) \oplus (c * a)$$ [Right Distribution]

Then operation * is said to be distributive ~~over~~ over the operation ⊕

Illustration :

Let R be the set of real number.

and since for all $a, b, c \in R$,

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$$ [Left Distribution]

& $$(b + c) \cdot a = (b \cdot a) + (c \cdot a)$$ [Right Distribution]

⇒ operation multiplication (·) is distributive ~~over~~ over the operation addition (+)

But since

$$a + (b \cdot c) \neq (a + b) \cdot (a + c) \quad \forall (a, b, c \in R$$

Similarly

$$(b \cdot c) + a \neq (b + a) \cdot (c + a)$$

⇒ operation addition (+) is not distributive over the operation multiplication (·).

Scanned by CamScanner

Some Algebraic Structures

* **Groupoid** : let G be the any non-empty set and '$*$' be the binary operation define on set G Then the algebraic structure $(G, *)$ is said to be Groupoid if it satisfy closure property. only.

  i.e. $\forall\ a, b \in G, \Rightarrow\ a * b \in G.$

* **Semigroup** : An algebraic structure $(G, *)$ is said to be a semigroup if it satisfy following properties

  - ① closure : $\forall\ a, b \in G,\quad a * b \in G$
  - ② Associative : $\forall\ a, b, c \in G,\quad a * (b * c) = (a * b) * c$

* **Monoid** :

  An algebraic structure $(G, *)$ is said to be Monoid if it satisfy following properties

  - ① closure : $\forall\ a, b \in G,\quad a * b \in G$
  - ② Associative : $\forall\ a, b, c \in G,$

    $$a * (b * c) = (a * b) * c$$

  - ③ Existance of Identity :

    $\forall\ a \in G,\quad \exists\ e \in G$ such that

    $$a * e = a = e * a$$

Scanned by CamScanner

✱ Group :

An algebraic structure $(G, *)$ is said to be a group. w.r.to operation $*$ if it satisfying following properties

(i) Closure: $\forall\ a, b \in G$, $a * b \in G$

(ii) Associative: $\forall\ a, b, c \in G$,

$$a * (b * c) = (a * b) * c$$

(iii) Existance of Identity

$\forall\ a \in G$, $\exists$ an element $e \in G$ such that

$$a * e = a \quad \text{OR} \quad e * a = a$$

(iv) Existance of Identity :

let $e \in G$ is identity w.r.t operation $*$ and $a \in G$ is any element of G. then for $\exists$ some $b \in G$ such that

$$a * b = e \Rightarrow b \text{ is inverse of } a$$

✱ Abelian Group

let $(G, *)$ be a group under the operation $*$ and it also satisfy Commutative property (ie $a * b = b * a$, $\forall\ a, b \in G$) then G is said to be abelian group w.r.to. operati '$*$'.

CAI

CAI

Id, Inv, Comm

Scanned by CamScanner

✱ Finite group & Infinite group

A group $(G, *)$ is called a finite group if the set G contain finite number of elements otherwise it is known as infinite Group.

✱ order of group :

The total number of elements in a finite group $(G, *)$ is the order of group G. and symbolically it is denoted as $|G|$ or $o(G)$

Ex : Let $G = \{1, \omega, \omega^2\}$ is set of cube root of unity. then show that G is an abelian group under multiplication operation

Soln : let $G = \{1, \omega, \omega^2\}$

Since G is set of cube root of unity

i.e. $\omega^3 = 1$ $\Rightarrow$ $\omega^4 = \omega^3 \cdot \omega = 1 \cdot \omega = \omega$

$\omega^5 = \omega^4 \cdot \omega = \omega \cdot \omega = \omega^2$

$\omega^6 = \omega^5 \cdot \omega = \omega^2 \cdot \omega = \omega^3 = 1$

and so on.

we prepear a composition table as follows

$\cdot$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

Scanned by CamScanner

From the composition table.

① ~~Set~~ clouser :

Since all entries in composition table are the elements of ~~of~~ set G.

$\Rightarrow$ G is closed w.r.to. multiplication

i.e. $\forall\ a, b \in G$, $a \cdot b \in G$.

for eg. $1, \omega^2 \in G \Rightarrow 1 \cdot \omega^2 = \omega^2 \in G$

② Associative :

Since $1, \omega, \omega^2 \in G$

and $1 \cdot (\omega \cdot \omega^2) = 1 \cdot \omega^3 = 1 \cdot 1 = 1$

& $(1 \cdot \omega) \cdot \omega^2 = \omega \cdot \omega^2 = \omega^3 = 1$

$\Rightarrow \boxed{1 \cdot (\omega \cdot \omega^2) = (1 \cdot \omega) \cdot \omega^2}$

set G. satisfying associative property w.r.to. multiplication operation.

③ Existance of identity

Since $1 \cdot \omega = \omega$
$1 \cdot \omega^2 = \omega^2$
$1 \cdot 1 = 1$
} (From composition table)

and also $1 \in G$. thus 1 is the multiplic

Identity of ~~Set~~ G.

④ Existance of Inverse :

From the composition table

Since $1 \cdot 1 = 1 \Rightarrow$ 1 is invers of itself

and $\omega \cdot \omega^2 = 1 = \omega^2 \cdot \omega \Rightarrow$ ω is inverse of ω^2 and vice versa

$\Rightarrow$ every element of set G have inverse in set G

Scanned by CamScanner

⑤ Commutative:

Since for any two element

$x, y \in G \Rightarrow x \cdot y = y \cdot x$

for eg:

$1 \cdot \omega^2 = \omega^2 = \omega^2 \cdot 1$

$\omega \cdot \omega^2 = \omega^2 \cdot \omega$

∴ It satisfy Commutative property w.r.to multiplication operation

∴ $(G, \cdot)$ is an abelian group

Ex: Prove that Set $G = \{1, -1, i, -i\}$ where $i = \sqrt{-1}$ is an abelian group under ordinary multiplication.

Soln: let $G = \{1, -1, i, -i\}$ where $i = \sqrt{-1}$

Composition table (w.r.t. operation multiplication)

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

① Closure: Since $\forall\ x, y \in G.$, $x \cdot y \in G.$

i.e. from table.

$(1) \cdot (-1) = -1 \in G$

$(-i) \cdot (-1) = i \in G$

$(-i) \cdot (i) = 1 \in G$ etc.

ie. all entries in the table are elements of set G

∴ G is closed w.r.to. operation multiplication.

Scanned by CamScanner

② Associative:

For any $x, y, z \in G$

since $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ is satisfied

$\therefore$ G is satisfying Associative property w.r.to operation multiplication.

for eg: For $1, i, -i \in G$

Sinc $1 \cdot (i \cdot -i) = 1 \cdot 1 = 1$

also $(1 \cdot i) \cdot -i = i \cdot -i = 1$

$\Rightarrow$ $\boxed{1 \cdot (i \cdot -i) = (1 \cdot i) \cdot -i}$

③ Existance of Identity:

from the composition table

since $1 \cdot 1 = 1$
$1 \cdot -1 = -1$
$1 \cdot i = i$
$1 \cdot -i = -i$

$\Rightarrow$ 1 is the identity of set G

$\because 1 \in G \Rightarrow$ Identity element exist in G

④ Existance of Inverse,

since $1 \in G$ is an identity element of set G w.r.to multiplication operation

and also. $1 \cdot 1 = 1 \Rightarrow$ 1 is inverse of itself

$-1 \cdot -1 = 1 \Rightarrow$ -1 is inverse of itself

$i \cdot -i = \Rightarrow$ i is inverse of -i and vice versa.

$\Rightarrow$ Inverse of each element of set G are exist in set G.

⑤ Commutative: Since for any $x, y \in G$

we have $x \cdot y = y \cdot x$

$\therefore$ Set G satisfying commutative property under multplication operation.

Scanned by CamScanner

for ex: let $1, -i \in G$

and $1 \cdot -i = -i$
& $-i \cdot 1 = -i$ $\Rightarrow (1)\cdot(-i) = (-i)\cdot(1)$

$\therefore$ G is commutative w.r.to. multiplication.

$\Rightarrow$ G is an abelian group under multiplication operation.

Ex: Prove that set of integers, I form an abelian group w.r. to operation $*$ which is defined as

for all $a, b \in I$, $a * b = a + b + 1$

Soln: Let I be the set of ~~operation~~. Integers and operation $*$ defined as

for all $a, b \in I$, $a * b = a + b + 1$ —①

① Closure: let $x, y \in I$ (are integers)

and $1 \in I$.

$\therefore$ $x + y + 1$ is also an integer.

$\Rightarrow$ $x * y \in I$

$\Rightarrow$ It is closed w.r.to operation $*$.

② Associative.

let $x, y, z \in I$

and $x * (y * z) = x * (y + z + 1)$ (By defn ①)

$= x + (y + z + 1) + 1$

$= x + y + z + 2$

& $(x * y) * z = (x + y + 1) * z$

$= x + y + 1 + z + 1 = x + y + z + 2$

$\Rightarrow \boxed{x * (y * z) = (x * y) * z}$ $\therefore$ It is associative.

Scanned by CamScanner

③ Existance of Identity

let $x \in I$ is any integer.

Such that $x * e = x$ (where e is identity element)

$\Rightarrow x + e + 1 = x$ (By defn)

$\Rightarrow e + 1 = 0$

$\Rightarrow e = -1 \in I$ ($\because -1$ is integer)

$\Rightarrow e = -1$ is an identity element of I

$\therefore$ Identity is exist..

④ Existance of Inverse :

we know that $e = -1 \in I$ is the identity element.

Now for any $x \in I$, some y

Such that $x * y = e = -1$

Then y is called inverse of x & viceversa.

Now $x * y = -1 \Rightarrow x + y + 1 = -1$ (By defn ①)

$\Rightarrow x + y = -2$

$\Rightarrow y = -2 - x$

$\Rightarrow y = -(2 + x)$

$\Rightarrow y = -(2+x)$ is inverse of x and

$y \in I$. ($\because 2, x \in I$

$\Rightarrow -(x+2) \in I$)

$\therefore$ Inverse of each $x \in I$ is exist.

⑤ Commutative

let $x, y \in I$

and $x * y = x + y + 1$

$= y + x + 1$

$= y * x$ $\Rightarrow$ It satisfy commutative property.

$\therefore$ Set of integer under operation * is abelian group.

Scanned by CamScanner

Ex: Prove that the set $\mathbb{Q}$ of rational numbers other than 1 form an abelian group with respect to operation $*$ define as

$\forall\ a, b \in \mathbb{Q}.\ (a \neq 1,\ b \neq 1,)$, $\underline{a * b = a + b - a \cdot b}$

Soln: let $\mathbb{Q}$ be the set of all rational numbers other than 1.

and operation $*$ define as.

$$\forall\ a, b \in \mathbb{Q}.\ (\text{with } a \neq 1,\ b \neq 1)$$

$$\Rightarrow\ a * b = a + b - a \cdot b \quad \text{——— (1)}$$

(1) Closure:

let $a \neq 1$, & $b \neq 1$ and $a, b \in \mathbb{Q}$

To prove closure property we need to show that

$$a * b \in \mathbb{Q} \ \&\ a * b \neq 1.$$

Now. since $a \in \mathbb{Q}$. & $b \in \mathbb{Q}$ also $a \neq 1, b \neq 1$

$\Rightarrow$ ~~$a + b \neq 1$~~ ~~& $ab \neq 1$~~

$\therefore\ (a + b) - ab \neq 1$ & Hence $a + b - ab \in \mathbb{Q}$

i.e. $a * b \in \mathbb{Q}$.

$\therefore$ It is closed w.r.to operation $*$ defined by eq^n (1).

because if we assume

$$(a + b) - ab = 1$$

$$\Rightarrow\ a + b - ab - 1 = 0$$

$$\Rightarrow\ (a - 1)(1 - b) = 0 \quad \Rightarrow a = 1 \ \&\ b = 1$$

This is contradiction

$\therefore$ our assumtion is wrong

$\Rightarrow \boxed{a + b - ab \neq 1}$ and it belong to $\mathbb{Q}$.

Scanned by CamScanner

② Associative :

let $a, b, c \in Q$ & $a \neq 1,\ b \neq 1\ c \neq 1$

Now. $a * (b * c) = a * [b + c - bc]$ (By eqn ①)

$= a + (b+c-bc) - a\cdot(b+c-bc)$

$= a+b+c-bc-ab-ac+abc$

& $(a*b)*c = [a+b-ab] * c$

$= (a+b-ab) + c - (a+b-ab)\cdot c$

$= a+b-c-ab-ac-bc+abc$

$\Rightarrow$ $\boxed{a*(b*c) = (a*b)*c}$

$\therefore$ It is Associative.

③ Existance of Identity :

let e is the identity element such that

$\forall\ a \in Q.$ $a * e = a$ & $a \neq 1$

we need to show that $e \in Q$ & $e \neq 0$

let $a * e = a \Rightarrow a + e - a\cdot e = a$

$\Rightarrow e - ae = 0$

$\Rightarrow e(1-a) = 0$

$\Rightarrow e = 0$ ($\because a \neq 1 \therefore 1-a \neq 0$)

Since $e=0$ is rational number & also $0 \neq 1$

$\Rightarrow e = 0 \in Q$ is identity element.

Scanned by CamScanner

④ Existance of Inverse.

we know that $e = 0 \in Q$ is identity element

let b is any element (any rational number)

such that $a * b = 0$ $\forall\ a \in Q$ & $a \neq 1$

Then we can say that b is inverse of a

Now we need to show that $b \neq 1$ & $b \in Q$

Since $a * b = 0 \Rightarrow a + b - ab = 0$

$\Rightarrow b(1-a) = -a$

$\Rightarrow b = \frac{a}{a-1}$

Sma $a \in Q$ & $a \neq 1$ $\Rightarrow \frac{a}{a-1} \neq 1$

$\therefore b = \frac{a}{a-1}$ is inverse of a and $b \in Q$.

⑤ Commutative :

let $a, b \in Q$ s.t $a \neq 1$, $b \neq 1$

Now $a * b = a + b - ab$

$= b + a - b \cdot a$ ($\because$ addition and multiplication is alway commutative)

$= b * a$

$\therefore$ It satisfy commutative property.

$\Rightarrow$ Q set of rational number other than 1 is form an abelian group under the operation * defined as $a * b = a + b - ab$.

Scanned by CamScanner

Ex: Prove that set Q_+ of positive rational number form an abelian group with respect to operation $*$ defined as $\forall\ a, b \in Q_+$, $a * b = \frac{a \cdot b}{2}$.

Soln: Let Q_+ be the set of positive rational number and operation $*$ is define as

$$a * b = \frac{a \cdot b}{2} \quad \forall\ a, b \in Q_+ \qquad \text{---(1)}$$

① **closure law.**

let $x, y \in Q_+ \Rightarrow x \cdot y \in Q_+$ [multiplication of Rational no. is always Rational]

also $2 \in Q_+$.

$\therefore \frac{(x \cdot y)}{2} \in Q_+$

$\Rightarrow x * y \in Q_+$.

$\therefore$ It is closed under operation $*$

② **Associative law.**

let $x, y, z \in Q_+$

$\therefore x * (y * z) = x * \left(\frac{y \cdot z}{2}\right)$ (∵ By defn ①)

$= \frac{\left(x \cdot \frac{(yz)}{2}\right)}{2} = \frac{xyz}{4}$

and $(x * y) * z = \left(\frac{x \cdot y}{2}\right) * z$

$= \frac{\left(\frac{x \cdot y}{2}\right) \cdot z}{2} = \frac{xyz}{4}$

$\Rightarrow x * (y * z) = (x * y) * z$

$\therefore$ It satisfy associative law w.r.to operation $*$

CARD

Scanned by CamScanner

③ Existance of Identity :

let $x \in Q_+$ be the any positive rational number

and we know that

$$x = x$$

$$\Rightarrow 2x = 2x$$

$$\Rightarrow \frac{2 \cdot x}{2} = x$$

$$\Rightarrow 2 * x = x \quad (\because \text{By defn } ①)$$

$\Rightarrow$ 2 is identity element and also belongs to Q_+.

④ Existance of Inverse :

We know that $e = 2 \in Q_+$ is an identity element of set Q_+ w.r.to operation $*$.

let b is any rational number such that $\forall a \in Q$,

$$a * b = e$$

ie. b is inverse of a

We need to show that $b \in Q_+$

Since $a * b = e = 2$

$$\Rightarrow \frac{a \cdot b}{2} = 2 \quad \because (\text{By defn } ①)$$

$$\Rightarrow b = 4/a$$

$$\Rightarrow b \in Q_+ \quad \because \left(a \in Q_+ \ \& \ 4 \in Q_+ \ \therefore 4/a \in Q_+ \right)$$

Thus $b = 4/a$ is an inverse of $a \in$ and also it belongs to Q_+.

⑤ Commutative property.

let $x, y \in Q_+$ & $x * y = \frac{x \cdot y}{2}$

$$= \frac{y \cdot x}{2} = y * x$$

$\therefore$ It is commutative.

$\Rightarrow$ Q_+ is an abelian group under the operation $*$.

Scanned by CamScanner

Ex: Show that set of matrices of the type

$$A_\alpha = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \quad \text{where } \alpha \in R \ (\text{Set of Real numbers})$$

form an abelian group w.r.to matrix multiplication.

Soln: let M be the set of matrices of the type

$$A_\alpha = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \quad \text{where } \alpha \in R. \qquad \text{---(1)}$$

ie. $M = \{ A_{\alpha_i} / \forall \alpha_i \in R, i = 1, 2, 3, \ldots \}$

① **Closure:** let $\alpha, \beta \in R$ (are any two real numbers)

and $A_\alpha = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}, A_\beta = \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix} \in M$

Now

$$A_\alpha \times A_\beta = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\alpha \cdot \cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -(\sin\alpha\cos\beta + \cos\alpha\sin\beta) \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & \cos\alpha\cos\beta - \sin\alpha\sin\beta \end{bmatrix}$$

$$A_\alpha \times A_\beta = \begin{bmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix} \qquad \text{---(2)}$$

Now since $\alpha, \beta \in R \Rightarrow \alpha+\beta \in R$ (addition of Real number is Real number)

$\Rightarrow A_\alpha \times A_\beta \in M$ (∵ By defn)

∴ It is closed under the matrix multiplication operation

Scanned by CamScanner

② Associative law

Let $\alpha, \beta, \gamma \in R$ and $A_\alpha, A_\beta, A_\gamma \in M$

where $A_\alpha = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, $A_\beta = \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix}$ & $A_\gamma = \begin{bmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{bmatrix}$

Now.

$$A_\alpha \times (A_\beta \times A_\gamma) = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \times \left\{ \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix} \times \begin{bmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{bmatrix} \right\}$$

$$= \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \times \begin{bmatrix} \cos(\beta+\gamma) & -\sin(\beta+\gamma) \\ \sin(\beta+\gamma) & \cos(\beta+\gamma) \end{bmatrix} \quad (\because \text{from eqn } ②)$$

$$= \begin{bmatrix} \cos(\alpha+\beta+\gamma) & -\sin(\alpha+\beta+\gamma) \\ \sin(\alpha+\beta+\gamma) & \cos(\alpha+\beta+\gamma) \end{bmatrix} \quad \forall\ \alpha+\beta+\gamma \in R.$$

$$= \begin{bmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix} \times \begin{bmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{bmatrix}$$

$$= \left\{ \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \times \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix} \right\} \times \begin{bmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{bmatrix}$$

$$= (A_\alpha \times A_\beta) \times A_\gamma$$

∴ It is associative w.r.to matrix multiplication.

③ Existance of Identity.

Since $0 \in R$ we can say that $A_0 = \begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} \in M$

Now.

$$A_\alpha \times A_0 = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \times \begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\alpha+0) & -\sin(\alpha+0) \\ \sin(\alpha+0) & \cos(\alpha+0) \end{bmatrix} \quad (\text{By eqn } ②)$$

$$= \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$\therefore A_\alpha \times A_0 = A_\alpha$$

Scanned by CamScanner

$\Rightarrow$ A_0 is identity matrix and it belongs to M.

④ Existance of Inverse

we know that $A_0 = \begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix}$ is

and identity element of set of matrices M.

Now. Since

$$A_0 = \begin{bmatrix} \cos 0 & -\sin 0 \\ \sin 0 & \cos 0 \end{bmatrix} = \begin{bmatrix} \cos(\alpha-\alpha) & -\sin(\alpha-\alpha) \\ \sin(\alpha-\alpha) & \cos(\alpha-\alpha) \end{bmatrix}$$

$\because$ for any $\alpha \in R$, $\alpha - \alpha = 0$

$$= \begin{bmatrix} \cos(\alpha+(-\alpha)) & -\sin(\alpha+(-\alpha)) \\ \sin(\alpha+(-\alpha)) & \cos(\alpha+(-\alpha)) \end{bmatrix}$$

$$= \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \times \begin{bmatrix} \cos(-\alpha) & -\sin(-\alpha) \\ \sin(-\alpha) & \cos(-\alpha) \end{bmatrix}$$

$\because$ Since $\alpha \in R \Rightarrow -\alpha \in R$

$\Rightarrow$ $A_0 = A_\alpha \times A_{(-\alpha)}$

$\Rightarrow$ $A_{(-\alpha)}$ Matrix is an inverse of matrix A_α and vice versa.

and also $A_{(-\alpha)} \in M$. as $(-\alpha) \in R$.

⑤ commutative.

let $A_\alpha, A_\beta \in M$ where $\alpha, \beta \in R$

we know that $A_\alpha \times A_\beta = \begin{bmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix}$; (from eqn ②)

$$= \begin{bmatrix} \cos(\beta+\alpha) & -\sin(\beta+\alpha) \\ \sin(\beta+\alpha) & \cos(\beta+\alpha) \end{bmatrix}$$ ($\because \alpha+\beta = \beta+\alpha$ addition is always commutative)

$$= A_\beta \times A_\alpha$$

$\therefore$ It is commutative w.r.t Matrix multiplication

$\Rightarrow$ Set $M = \{[A_{\alpha_i}] \mid \alpha_i \in R\}$ is an abelian group.

Scanned by CamScanner

~~operations~~ Some other operations

* Addition modulo m :

let a and b are any two distinct integers and m is fixed positive integer then addition modulo m is denoted by $a +_m b$, which is define as follow.

$$a +_m b = r, \quad \text{where } \boxed{0 \le r < m}$$

Where r is the least non-negative remainder when (a+b) is divided by m.

for ex :

① $14 +_7 8 = 1$

∵ $14+8 = 22$
& If 22 divided by 7 then least remainder is 1 ∴ $22 = 7(3)+1$

② $(-19) +_4 5 = 2$

since $-19+5 = -14$
& If -14 is divided by 4 then remainder is 2 (≥ 0) as $-14 = 4(-4)+2$

* multiplication modulo m

let a and b are any two integers and m is fixed positive integer. then multiplication modulo m is denoted and define as

$$a \times_m b = r, \quad \text{where } \boxed{0 \le r < m}$$

where, r is the least non-negative ~~integer~~ remainder when (a*b) is divided by m.

for eg, ① $8 \times_5 3 = 4$

∵ $8 \times 3 = 24$ is divided by 5 then remainder $r = 4$ ∵ $24 = 5(4)+4$

② $-8 \times_5 3 = 1$

since $(-8)\times(3) = -24$ is divided by 5 then remainder is

$-24 = 5(4) - 4$
$= 5(4) - 5 + 5 - 4 = 5(-5) + 1$

∴ $-24 = 5(-5)+1$

Scanned by CamScanner

★ Congruence Relation

Let a and b are any two integers and m is fixed positive integer then the relation $a \equiv b \pmod m$ is said to be congruence relation and it is read as "a congruence to b mod.m"

It means, when (a-b) is divided by m we get non-negative remainder r which is actual value of $a \equiv b \pmod m$.

i.e. $[a \equiv b \pmod m] \Rightarrow a - b = r$, where $0 \le r < m$

for ex:

① $5 \equiv 2 \pmod 2 \Rightarrow 5 - 2 = 1$

∵ (5-2) = 3 and if we divide 3 by 2 then remainder r = 1.

②

② $14 \equiv 4 \pmod 3$

$\Rightarrow 14 - 4 = 1$

∵ 14-4 = 10 and if we devide 10 by 3 then remainder is r = 1

Scanned by CamScanner

Ex: Prove that the set $G = \{0, 1, 2, 3, 4, 5\}$ is a finite abelian group with respect to operation "addition modulo 6"

Soln: let $G = \{0, 1, 2, 3, 4, 5\}$ is finite set and given operation is "addition modulo 6" which can be define as $\forall\ a, b \in G,\quad a +_6 b = r$

where $0 \le r < m$, r is remainder obtain when we divide $(a+b)$ by 6.

∴ Composition table:

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

① Closure property:

Since all the entries in composition table are again elements of set G, it is closed w.r.to addition modulo 6.

ie $\forall\ a, b \in G,\ (a +_6 b) \in G$.

for eg: For $2, 4 \in G$, $2 +_6 4 = 0 \in G$.

Here $2, 3 \in G$, $2 +_6 3 = 5 \in G$.

② Associative property:

Since For any elements such as $a, b, c \in G$

$$a +_6 (b +_6 c) = (a +_6 b) +_6 c$$

∴ It is associative w.r.to. "addition modulo 6"

for ex:

$1 +_6 (3 +_6 5) = 1 +_6 2 = 3$

$(1 +_6 3) +_6 5 = 4 +_6 5 = 3$

$\Rightarrow 1 +_6 (3 +_6 5) = (1 +_6 3) +_6 5$

Scanned by CamScanner

③ Existance of Identity :

from the composition table

since $\forall\ a \in G, \quad a +_6 0 = a \in G$

$\therefore$ 0 is identity of set G w.r.to operation "addition modulo 6"

forex : $0 +_6 0 = 0$

$1 +_6 0 = 1$

$2 +_6 0 = 2$

$3 +_6 0 = 3$

$4 +_6 0 = 4$

$5 +_6 0 = 5$

④ Existance of Inverse :

Since $0 \in G$ is an identity element of set G w.r.to to "addition modulo 6" operation.

and from the table

$0 +_6 0 = 0 \Rightarrow$ 0 is inverse of itself.

$1 +_6 5 = 0 \Rightarrow$ 1 is inverse of 5 and vice versa

$2 +_6 4 = 0 \Rightarrow$ 2 is inverse of 4 & vice versa

$3 +_6 3 = 0 \Rightarrow$ 3 is inverse of itself.

$\Rightarrow \forall\ a \in G, \exists$ an e.invere $b \in G$. Such that

$a +_6 b = 0$ (identity)

$\therefore$ inverse of each elements of set G are exist in G.

⑤ Commutative property :

since from the composition table

$\forall\ a, b \in G, \quad a +_6 b = b +_6 a \Rightarrow$ It is commutative

forex : ① $3 +_6 4 = 1 = 4 +_6 3$ etc.

$\therefore$ Since set G satisfy closure, Associative, identity, inverse and commutative property w.r.to "addition modulo 6"

$\Rightarrow$ G is an abelian group w.r.to addition modulo 6.

Scanned by CamScanner

(H.W)

Ex (1) Prove that set $G = \{1, 2, 3, 4\}$ is finite abelian group with respect to operation multiplication modulo 5"

Hint: Composition table

$\times_5$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Ex: (2) Show that set of matrices.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

is form finite abelian group w.r.to operation matrix multiplication

Hint: Consider $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $D = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

and form a composition table w.r.to matrix multiplication as follows

$\times$	A	B	C	D
A	A	B	C	D
B	B	A	D	C
C	C	D	A	B
D	D	C	B	A

Ex (3) Prove that set of all matrices of the type $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ where $a, b \in R$ (Real number) is an abelian group under operation matrix ~~multiplication~~ addition

Ex (4) Show that set of all elements of the type $(a + \sqrt{2}\, b)$ where $a, b \in Q$ (Rational number) is form an abelian group w.r.to. matrix addition.

Scanned by CamScanner

✱ Some Important properties of Group

⑥ Let us consider Set G is group with respect to operation '$*$'

ie $(G, *)$ is a group.

Then it satisfy following properties.

① Closure property: $\forall\ a, b \in G, \Rightarrow a * b \in G$

② Associative property: for $a, b, c \in G$

$$a * (b * c) = (a * b) * c$$

③ Identity: $\exists$ an element $e \in G$ such that

$$\forall\ a \in G, \quad a * e = a = e * a$$

and e is known as identity element of G

④ Inverse: $\exists$ an element $a^{-1} \in G$ such that

for $a \in G$, $a * a^{-1} = e$ (where e is identity)

and Hence $a^{-1} \in G$ is the inverse of $a \in G$

⑤ Cancellation law:

let $(G, *)$ is a group.

then for $a * b = a * c \Rightarrow b = c \quad \forall a, b, c \in G$

[left cancellation law]

also $b * a = c * a \Rightarrow b = c$

[Right cancelation law]

Scanned by CamScanner

⑥ Inverse of inverse of any element of group $(G, *)$ is the element it self

ie $\forall\ a \in G$, (where G is group)

$\Rightarrow (a^{-1})^{-1} = a$

⑦ Reverse law

$\forall\ a, b \in G$, & G is group w.r.to operation $*$

Then $(a * b)^{-1} = b^{-1} * a^{-1}$

$\neq a^{-1} * b^{-1}$ (need not be true for all operations.)

ie. (Inverse of product of two elements of group G is the product of their inverses in reverse order)

Ex: Show that if every element of a group $(G, \circ)$ be it's own inverse then it is an abelian group.

Soln: Let $(G, \circ)$ is a group w.r.to operation '$\circ$'

and it is given that, every element of Group G are their own inverses.

ie. for $a \in G$, $a = a^{-1}$ (ie inverse of $a = a$)

for $b \in G$, $b = b^{-1}$ (i.e inverse of $b = b$) ——— (1)

Now; we know that

For any $a, b \in G \Rightarrow a \circ b \in G$ [Closure Property]

$\therefore$ By using (1) $\Rightarrow a \circ b = (a \circ b)^{-1}$

$\Rightarrow a \circ b = b^{-1} \circ a^{-1}$ (Reversal law)

$\Rightarrow a \circ b = b \circ a$ ($\because$ from (1))

$\therefore$ It satisfy commutative property. H.P

Scanned by CamScanner

$\therefore$ Group $(G, \circ)$ is an abelian group.

Ex: Show that, If a, b are arbitrary elements of group G, then $(a \circ b)^2 = a^2 \circ b^2$ iff G is abelian group.

Soln: let $(G, \circ)$ be a group w.r.to to operation $\circ$.

and let $\forall\ a, b \in G$, $(a \circ b)^2 = a^2 \circ b^2$ — (1) (ordinary multiplication)

To show G is abelian group

we need to show that it satisfy commutative property.

Now. let $(a \circ b)^2 = a^2 \circ b^2$ (given (1))

$\Rightarrow (a \circ b) \cdot (a \circ b) = (a \cdot a) \cdot (b \cdot b)$

$\Rightarrow a \circ (b \cdot (a \cdot b)) = a \cdot (a \cdot (b \cdot b))$ ($\because$ Associative property)

$\Rightarrow b \cdot (a \cdot b) = a \cdot (b \cdot b)$ $\because$ [left cancelation law]

$\Rightarrow (b \cdot a) \cdot b = (a \cdot b) \cdot b$ $\because$ [Associative property]

$\Rightarrow (b \cdot a) = a \cdot b$ $\because$ [Right cancelation law]

$\Rightarrow G$ satisfy commutative property

$\therefore$ group G is an abelian group

Conversly, let G is an abelian group

$\therefore\ a \circ b = b \circ a$ $\forall\ a, b \in G$ — (2)

We need to prove that

$(a \circ b)^2 = a^2 \circ b^2$ $\forall\ a, b \in G$

Scanned by CamScanner

let L.H.S = $(a \circ b)^2$

$= (a \circ b) \cdot (a \circ b)$

$= a \cdot ((b \cdot a) \cdot b)$ (∵ associative property)

$= a \cdot (a \cdot b) \cdot b$ [commutative property

$= (a \circ a) \cdot (b \circ b)$ (∵ Associative)

$\Rightarrow (a \circ b)^2 = a^2 \cdot b^2$

H.P.

---

* Order of an Element :⇒

let $(G, *)$ be a group w.r.to operation $*$ and $a \in G$ is any arbitary element of group G.

Then, the smallest positive integer n is said to be an order of element a if it satisfy condition

$$\boxed{a^n = \underbrace{a * a * a * \ldots * a}_{n \text{ times}} = e}$$

where e is identity of group G.

and it is denoted as $o(a) = n$

Note : An order of an identity element of every group G. is always 1

ie. If $e \in G$ is identity element of Group G

Then $o(e) = 1$

Scanned by CamScanner

Ex: Let $G = \{1, -1, i, -i\}$ be a group w.r.t to multiplication operation $\cdot$, then find the order of each element of group G.

Soln: Let $G = \{1, -1, i, -i\}$ is a group w.r.t to ordinary multiplication.

$\therefore$ Composition table.

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

From the table since,

$1 \cdot 1 = 1$
$-1 \cdot 1 = -1$
$i \cdot 1 = i$
$-i \cdot 1 = -i$

$\Rightarrow$ 1 is identity of Group G i.e $e = 1 \in G$.

① **order of 1:**

Since $1 \in G$ is an identity of group G

$\Rightarrow o(1) = 1$ [order of identity is always unity]

② **order of -1**

Since $(-1)^2 = 1$ $\quad \because (-1)^n \neq 1$ if n is odd

But $(-1)^n = 1$ if n is even

$\therefore n = 2, 4, 6, 8, 10, \ldots$

Since least value of $n = 2$

This shows that $(-1)^n = (-1)^2 = 1$

Scanned by CamScanner

$\therefore o(-1) = 2$ [ie order of (-1) is 2]

③ order of i

since $(i)^2 = i \cdot i = i^2 = -1 \neq 1$

$(i)^3 = i \cdot i \cdot i = (i \cdot i) \cdot i = (-1) \cdot i = -i \neq 1$

BW $(i)^4 = (i \cdot i)^2 \cdot (i)^2 = (-1) \cdot (-1) = 1$

$\Rightarrow$ $o(i) = 4$ i.e order of i is 4

④ order of $(-i)$

since $(-i)^2 = (-i) \cdot (-i) = (i \cdot i) = -1 \neq 1$

$(-i)^3 = (-i) \cdot (-i)^2 = (-i)(-1) = i \neq 1$

& $(-i)^4 = (-i)^2 \cdot (-i)^2 = (-1) \cdot (-1) = 1$

$\Rightarrow$ $o(-i) = 4$ i.e order of $(-i)$ is 4

Ex: let $G = \{0, 1, 2, 3\}$ is a group w.r.to operation, addition modulo 4, then find the order of each element of group G.

Soln: let $(G, +_4)$ is a group w.r.to addition modulo 4 where $G = \{0, 1, 2, 3\}$

∴ Composition table

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Scanned by CamScanner

from composition table,

since $\left.\begin{array}{l} 0 +_4 0 = 0 \\ 1 +_4 0 = 1 \\ 2 +_4 0 = 2 \\ 3 +_4 0 = 3 \end{array}\right\} \Rightarrow 0 \in G$ is identity element of group G.

Now,

① order of 0 :

Since $0 \in G$ is identity element

$\Rightarrow o(0) = 1$ $[(0)^1 = 0]$

i.e. order of 0 is 1

② order of 1 :

since $(1)^2 = 1 +_4 1 = 2 \neq 0$

$(1)^3 = 1 +_4 1 +_4 1 = 2 +_4 1 = 3 \neq 0$

$(1)^4 = (1 +_4 1) +_4 (1 +_4 1) = 2 +_4 2 = 0$

$\Rightarrow o(1) = 4$ i.e. order of 1 is 4

③ order of 2

since $(2)^2 = 2 +_4 2 = 0$

$\therefore o(2) = 2$ i.e order of 2 is 2

④ order of 3

Since $(3)^2 = 3 +_4 3 = 2 \neq 0$

$(3)^3 = (3 +_4 3) +_4 3 = 2 +_4 3 = 1 \neq 0$

But $(3)^4 = (3 +_4 3) + (3 +_4 3) = 2 +_4 2 = 0$

$\Rightarrow o(3) = 4$ i.e order of 3 is 4

Scanned by CamScanner

✱ Subgroup :

Defn : Let $(G, *)$ is a group w.r.to operation $*$, and H is any non-empty set.
Then an algebraic structure $(H, *)$ is said to be a subgroup of group G if it satisfy following ~~proof~~ conditions

(i) $(H, *)$ is group under the operation $*$

(ii) H is non-empty subset of set G. $[H \subseteq G]$

for eg : (1) Additive Group of even integer is a subgroup of additive group of all integers

(2) $(G, *)$ & $(\{e\}, *)$ are trivial subgroup of group G.

Thm : The identity of subgroup is same as identity of group.

Proof : Let $(H, *)$ is subgroup of group $(G, *)$.

i.e. $(H, *)$ is group w.r.to operation $*$ & $H \subseteq G$ } (∵ By defn)

Now let us consider $e \in G$ is identity of group G
& $e' \in H$ is identity of group H

we need to show that $e = e'$

let a be the any arbitary element. such that

$a \in H \Rightarrow a \in G$ (∵ $H \subseteq G$)

$\Rightarrow a * e \in G$ (closure property of group.)

Scanned by CamScanner

$\Rightarrow \boxed{a = a * e}$ ($\because e \in G$ is identity)

—①

Now. Since $e' \in H$ is identity of group H

and $\forall a \in H$, $\boxed{a * e' = a}$ —②

From ① & ②

$$a * e' = a * e$$

$\Rightarrow e' = e$ [cancellation law]

$\Rightarrow$ identity of subgroup H & group G are same.

Thm : A non-empty subset H of group $(G, *)$ is a subgroup of G iff $\forall a \in H, b \in H \Rightarrow a * b^{-1} \in H$

proof : $\Rightarrow$ Necessary part :

let $(H, *)$ is a subgroup of group G

$\Rightarrow$ (i) $H \subseteq G$ [H is non empty subset of set G]

(ii) $(H, *)$ is group w.r.to. operation $*$

To prove : $\forall a \in H, b \in H \Rightarrow a * b^{-1} \in H$

let $a \in H, b \in H \Rightarrow a, b \in G$ ($\because H \subseteq G$)

$\Rightarrow a^{-1}, b^{-1} \in H$ $\because$ (H is group & Hence inverse of each element must exist)

Scanned by CamScanner

Now since $a \in H$, & $b^{-1} \in H$.

$\Rightarrow a * b^{-1} \in H$ ∵ (By closure property of group)

Hence $\boxed{\forall\ a, b \in H \Rightarrow a * b^{-1} \in H}$ proved.

Conversly : [Sufficient part]

Let, $\forall\ a, b \in H \Rightarrow a * b^{-1} \in H$ ——① given.

To prove : H is subgroup of Group G

Since it is given that H is non-empty subset of Set G

$\Rightarrow H \subseteq G$

∴ To prove H is subgroup, we need to prove that $(H, *)$ is itself a group.

① Identity property :

let $\forall, a, b \in H \Rightarrow a * b^{-1} \in H$ ∵ (given ①)

Thus for $a \in H$, $a * a^{-1} \in H$ ——②

$\Rightarrow a * a^{-1} \in G$ ∵ $H \subseteq G$

and since a^{-1} is inverse of a,

$a * a^{-1} = e$ identity.

∴ from ② $e \in H$ i.e. identity is exist in H.

② Inverse property.

let $a \in H$ any arbitary element of H

& since $e \in H$ is identity of H

∴ $\forall\ a \in H, e \in H, \Rightarrow a * a^{-1} \in H$ (By eqn ①)

Scanned by CamScanner

$\Rightarrow e * a^{-1} \in G$ ($\because H \subseteq G$)

$\Rightarrow$ and G is group.

$\Rightarrow a * a^{-1} \in H$ $\because a^{-1}$ is inverse of a

$\Rightarrow a^{-1} \in H$

$\therefore$ Inverse of $a \in H$ is exist in H.

③ closure property:

let $\forall\ a, \in H$ & $b \in H$. $\Rightarrow a \in H,\ b^{-1} \in H$

($\because$ Inverse exis

$\Rightarrow a * (b^{-1})^{-1} \in H$ (By given eq^{n} ①)

$\Rightarrow$ $\Rightarrow a * b \in H$ ($\because (b^{-1})^{-1} = b$)

$\Rightarrow \therefore \forall\ a, b \in H \Rightarrow a * b \in H$.

$\therefore$ It satisfy closure property.

④ Associative property

let $a, b, c \in H$ & since $H \subseteq G$

$\Rightarrow a, b, c \in G$ & G is group

$\therefore$ It satisfy associative property

i.e $a * (b * c) = (a * b) * c$

$\Rightarrow \forall\ a, b, c \in H \Rightarrow a * (b * c) = (a * b) * c$

$\therefore$ Associative property Holds in H.

$\therefore$ $(H, *)$ is group.

Now, since $H \subseteq G$ & H itself is group

$\Rightarrow (H, *)$ is subgroup of group G.

Scanned by CamScanner

Thm: The intersection of any two subgroup of group $(G, *)$, is again a subgroup of group G.

Soln: Let H_1 & H_2 are ~~two distinct~~ any two subgroup of group G

and let $e \in G$ is an identity element of group G

We know that identity of subgroup is same as identity of group.

$\therefore$ e is also an identity of both subgroups H_1 & H_2

ie $e \in H_1$ & $e \in H_2$

$\Rightarrow$ $H_1 \cap H_2 \neq \phi$ [ie. set $H_1 \cap H_2$ is nonempty subset of group G.

ie $H_1 \cap H_2 \subseteq G$

Now, let $a, b \in (H_1 \cap H_2)$

$\Rightarrow a, b \in H_1$ and $a, b \in H_2$

$\Rightarrow a * b^{-1} \in H_1$ and $a * b^{-1} \in H$ $\because$ (By well known thm)

$\Rightarrow a * b^{-1} \in (H_1 \cap H_2)$

$\therefore$ By well known thm.

Since $\forall\ a, b \in (H_1 \cap H_2) \Rightarrow a * b^{-1} \in (H_1 \cap H_2)$

$\Rightarrow H_1 \cap H_2$ is a subgroup of group G

Scanned by CamScanner

Thm: If H_1 & H_2 are subgroup then $H_1 \cup H_2$ need not to be a subgroup of group G.

Soln: Let H_1 & H_2 are any two subgroup of group $(G, *)$ and since identity element of ~~group~~ subgroup is same as identity of group.

$\therefore$ let $e \in G$ is identity element of group G

$\Rightarrow e \in H_1$ & $e \in H_2$ ~~is~~ is identity of subgroup H_1 & H_2

i.e. $e \in H_1 \cup H_2$ is identity of set $H_1 \cup H_2$

i.e. $H_1 \cup H_2$ is non-empty subset of group G

To show $H_1 \cup H_2$ need not be a subgroup

we need to show that

for $a, b \in H_1 \cup H_2 \Rightarrow a * b^{-1} \notin H_1 \cup H_2$

let ~~$a, b \in (H_1 \cup H_2)$~~

$\Rightarrow$ ~~$a \in H_1$, $b \in H_1$ OR $ab \in H_2$~~

for $a \in H_1 \cup H_2 \Rightarrow a \in H_1$ OR $a \in H_2$

& for $b \in H_1 \cup H_2 \Rightarrow b \in H_1$ OR $b \in H_2$

Now consider the case

$a \in H_1$ & $b \notin H_1 \Rightarrow b^{-1} \notin H_1$

$\Rightarrow a * b^{-1} \notin H_1$ —①

Similarly if $a \in H_2$ & $b \notin H_2 \Rightarrow a * b^{-1} \notin H_2$ —②

$\therefore$ From eqn ① & ②

for ~~$a, b \in H$~~ $a * b^{-1} \notin (H_1 \cup H_2)$

ie for $a, b \in (H_1 \cup H_2) \Rightarrow a * b^{-1} \notin (H_1 \cup H_2)$

$\therefore$ $H_1 \cup H_2$ need not be a subgroup of group G.

Scanned by CamScanner

★ Coset :

Defn: let $(H, *)$ be a subgroup of group $(G, *)$ & for any $a \in G$, the set aH define as

$$aH = \{ a*h \ / \ h \in H, a \in G \}$$

is called as left coset of H in G, determine by element $a \in G$.

Here a is known as a representative of a left coset aH.

Similarly: A set $Ha = \{ h*a \ / \ h \in H, a \in G \}$ is called as right coset of H in G. determined by element $a \in G$.

★ Properties of coset [~~Right~~ Left coset]

Let H be a subgroup of group G.

and $a, b \in G$. then

(i) $a \in aH$

(ii) $aH = H \iff a \in H$

(iii) either $aH = bH$. OR $aH \cap bH = \phi$

(iv) $aH = bH$ iff $a^{-1} \cdot b \in H$.

Note: we can define similar properties for Right cosets

Scanned by CamScanner

Thm: Any two right cosets of subgroup H are either identical or disjoint.

OR

let H be a subgroup of group G and aH & bH are right cosets of subgroup G respectively determine by $a \in G$ & $b \in G$.

Then $aH = bH$ **OR** $aH \cap bH = \phi$

proof: let H is a subgroup of group G

and aH and bH are two right cosets of subgroup determine by element $a \in G$, & $b \in G$ respectively.

let us assume that $aH \cap bH \neq \phi$.

$\Rightarrow$ $\exists$ some element $x \in (aH \cap bH)$

$\Rightarrow$ $x \in aH$ & $x \in bH$

$\Rightarrow$ $x = a*h$ & $x = b*h$ (For $h \in H$)

$\Rightarrow$ $a*h = b*h$ $\because$ (By defn of coset)

$\Rightarrow$ $a = b$ ($\because$ By cancellation law)

Since a & b are same

$\therefore$ $\boxed{aH = bH}$ H.P.

Similarly:

assume that $aH \neq bH$

($\because$ ~~any two distinct cosets~~)

$\Rightarrow$ $a \neq b$

$\Rightarrow$ $a*h \neq b*h$ $\forall$ $h \in H$

Since $a*h \in aH$ & $b*h \in bH$ (By defn)

$\Rightarrow$ aH and bH are two distinct sets

i.e $\boxed{aH \cap bH = \phi}$ (Disjoint sets)

H.P.

Scanned by CamScanner

* Normal subgroup :

A subgroup $(H, *)$ of group $(G *)$ is said to be a normal subgroup. if right coset of H generated by any element $a \in G$ is same as left coset of H generated by same element $a \in G$.

ie. If $aH = Ha \quad \forall a \in G$

then H is called as Normal subgroup

Thm: A subgroup H of a group G is a normal subgroup if and only if $(g^{-1} * h * g) \in H \quad \forall h \in H, g \in G$.

Proof: Let us consider,

A subgroup H of group G is normal subgroup.

i.e $\forall g \in G$, $gH = Hg$ —— (1) (∵ By defn)

Now: we show that $\forall h \in H$, & $g \in G$, $(g^{-1} * h * g) \in H$

let $h \in H$ & $g \in G \Rightarrow h * g \in Hg$ [By defn of right coset]

$\Rightarrow h * g \in gH$ (∵ By eqn (1)) —— (2)

we also know that, since G is group.

and $g \in G \Rightarrow g^{-1} \in G$ [Inverse is exist]

~~Now, let us consider an element $h_1 \in H$~~

Now, by defn of left coset

for any $h_1 \in H$, & $g \in G$, we have

$g * h_1 \in gH$ —— (3)

Scanned by CamScanner

Now it is obvious that

Since $h*g \in gH$ and $g*h_1 \in gH$ (from ② & ③)

for some $h_1 \in H$, $\quad h*g = g*h_1$

$\Rightarrow g^{-1}*(h*g) = g^{-1}*(g*h_1)$ $\quad [\because g^{-1} \in G]$

$\Rightarrow g^{-1}*h*g = (g^{-1}*g)*h_1$

$= e*h_1$ $\quad$ ($\because$ e is identity)

$\Rightarrow g^{-1}*h*g = h_1 \in H$ $\quad \forall\, g \in G,\ \&\ h \in H$

$\therefore$ Hence it is prove that. If H is normal subgroup of group G. then $g^{-1}*h*g \in H$

Now. Converly,

let H be a subgroup, such that

$g^{-1}*h*g \in H,\ \forall\ h \in H,\ g \in G.$ ——④

we show that H is normal subgroup

ie. we need to show that $\forall\, g \in G.\ gH = Hg$

Since $h \in H$, & $g \in G \Rightarrow h*g \in Hg$ ($\because$ By defn of Right coset)

Now. Since.

$(g^{-1}*h*g) \in Hg \Rightarrow h*g = g*e*(h*g)$

($\because$ e is identity)

$\Rightarrow h*g = (g*g^{-1}).(h*g)$

$= g*(g^{-1}*h*g)$ (∵Associative)

$\Rightarrow h*g \in gH$

$\Rightarrow \boxed{Hg \subseteq gH}$ ——⑤

[←] $g^{-1}*h*g \in H$ & $g \in G$ ∴ By defn of left coset]

Scanned by CamScanner

Similarly let, $b = g^{-1} \in G$

Using eqn (4), since $b \in G$ & For $h \in H$,

$$\Rightarrow b^{-1} * h * b \in H \quad (\because H \text{ is normal subgroup})$$

$$\Rightarrow (g^{-1})^{-1} * h * (g^{-1}) \in H$$

$$\Rightarrow g * h * g^{-1} \in H \quad \text{---(6)}$$

Now: $\forall\ h \in H$, and $g \in G \Rightarrow g * h \in gH$ [$\because$ By defn of left coset]

For $(g*h) \in gH$

we have $g*h = (g*h) * e$ [$\because$ e is identity $= a*a^{-1}$]

$$= (g*h) * (g^{-1} * g)$$

$$= (g*h*g^{-1}) * g$$

$\Rightarrow g*h \in Hg$ $\quad \forall.\ g \in G$ & $g*h*g^{-1} \in H$

$\Rightarrow \boxed{gH \subseteq Hg}$ ---(7)

Now from (5) & (7), since $Hg \subseteq gH$

& $gH \subseteq Hg$

$\Rightarrow \boxed{Hg = gH}$

i.e. H is Normal subgroup of group G.

Scanned by CamScanner

Ex: Show that the intersection of any two Normal subgroup of group G is also a normal subgroup.

Soln: let H_1 & H_2 are any two normal subgroup of a group G

we show that: $H_1 \cap H_2$ is also a normal subgroup.

If H_1 is normal subgroup then by previous thm. $\forall\ g \in G$ & $h \in H_1$, $g^{-1} * h * g \in H_1$ —①

Now let us consider x be the any arbitary element of $H_1 \cap H_2$

i.e. $x \in H_1 \cap H_2 \Rightarrow x \in H_1$ & $x \in H_2$

Now for any element $g \in G$, & $x \in H_1$

Since H_1 is Normal

$\therefore$ $\boxed{g^{-1} * x * g \in H_1}$ (By eqn ①)

Similarly for any element $g \in G$ & $x \in H_2$

Since H_2 is Normal. $\boxed{g^{-1} * x * g \in H_2}$ (∵ By de ①)

$\Rightarrow \forall\ x \in H_1 \cap H_2$ & $g \in G$, we have

$$g^{-1} * x * g \in (H_1 \cap H_2)$$

$\Rightarrow$ $H_1 \cap H_2$ is a Normal subgroup.

Scanned by CamScanner

Ex: Show that every subgroup of an abelian group is Normal.

Soln: Let H be the any subgroup of abelian group.

To show: H is Normal subgroup

we show that $\forall\ x \in H,\ g \in G \Rightarrow g^{-1} * x * g \in H$.

Now, let $x \in H$ & $g \in G$

$$\Rightarrow g^{-1} * x * g = g^{-1} * (x * g) \quad (\because \text{Associative})$$

$$= g^{-1} * (g * x) \quad (\because \text{commutative}\ \because G \text{ is abelian})$$

$$= (g^{-1} * g) * x \quad (\text{Associative})$$

$$= e * x \quad [\text{identity}]$$

$$\Rightarrow \boxed{g^{-1} * x * g = x} \quad \forall\ x \in H$$

$$\Rightarrow g^{-1} * x * g \in H.$$

$\therefore$ H is Normal subgroup.

Scanned by CamScanner

Note: Some defmition of function's

① one-one function: (Injective)

A function $f: A \longrightarrow B$ is said to be one-one or injective function.

if for $x_1, x_2 \in A$ (co-domain) such that

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

② onto function: (surjective)

A function $f: A \longrightarrow B$ is said to be on-to (OR surjective) function. if for every element $y \in B$ (codomain) $\exists$ some element $x \in A$ (Domai

such that $y = f(x)$

③ one-one & onto [Bijective]

A function $f: A \longrightarrow B$ is said to be Bijective if it is one-one and onto both.

Scanned by CamScanner

★ Homomorphism :⇒

Let $(G_1, *)$ and $(G_2, \oplus)$ be any two group and let $f: G_1 \longrightarrow G_2$ is a function defined as

$$f(a*b) = f(a) \oplus f(b) \quad \forall\ a, b \in G_1$$

$f(a), f(b) \in G$.

then f is called group Homomorphism.

★ Isomorphism :⇒

If a function $f: (G_1, *) \longrightarrow (G_2, \oplus)$ is one-one and on to and also f is Homomorphism then function f is called as Isomorphism betn G_1 & G_2 and G_1 & G_2 are said to be isomorphic

ie. $G_1 \cong G_2$

Scanned by CamScanner

Ex: If $G=\{1, i, -1, -i\}$ is group under multiplication and $G'=\{0, 1, 2, 3\}$ is group under addition modulo 4 then show that G & G' are isomorphic.

Soln: Let $G=\{1, i, -1, -i\}$ $\quad$ $G'=\{0, 1, 2, 3\}$

$\cdot$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Let $f : (G, \cdot) \longrightarrow (G'', +_4)$

from table $\quad f(1)=0,\ f(i)=1,\ f(-1)=2,\ f(-i)=3$ } f is one-one

Also, for every $y \in G'$ $\exists$ some $x \in G$ s.t. $y = f(x)$

$\therefore$ f is onto

Now, $\quad f(a \cdot b) = f(a) +_4 f(b) \quad \forall\ a, b \in G$

For eg. $\quad f(1 \cdot -i) = f(-i) = 3 \quad \longrightarrow *$

& $f(1) +_4 f(-i) = 0 +_4 3 = 3$

$\therefore$ $f(1 \cdot -i) = f(1) +_4 f(-i) = 0 +_4 3 = 3$

$\therefore$ f is Homorphism.

$\Rightarrow$ G is isomorphic to G'

$\Rightarrow$ $G \cong G'$

Scanned by CamScanner

Ex: let E be the set of all even integers. Show that semigroup $(Z, +)$ & $(E, +)$ are isomorphic. where Z is set of +ve integers

Soln: let $f: (Z, +) \longrightarrow (E, +)$ is a function which can be define as

$$\boxed{f(x) = 2x}, \quad \forall\, x \in Z \qquad \text{---(1)}$$

[x is any +ve integer]
$\therefore$ 2x is even integer.

To show f is one-one :⇒

let, x_1 & x_2 are any two +ve integers

i.e. $x_1, x_2 \in Z$

Such that $x_1 \neq x_2 \Rightarrow 2x_1 \neq 2x_2$

$\Rightarrow f(x_1) \neq f(x_2)$ ($\because$ By defn (1))

$\Rightarrow$ f is one-one.

To show f on-to :⇒

let y be any arbitary even integer

i.e $y \in E \Rightarrow$ y is divisible by 2

$\therefore$ for any $x \in Z$, $\frac{y}{2} = x$

$\Rightarrow y = 2x$

$\Rightarrow y = f(x)$

$\Rightarrow \forall\, y \in E \;\; \exists$ some $x \in Z$ such that $y = f(x)$

$\therefore$ f is onto.

Since f is one-one and onto $\Rightarrow$ f is Bijective function

Now To show $f: (Z, +) \longrightarrow (E, +)$ is Homomorphism

Here $(Z, +)$ & $(E, +)$ are semigroup w.r.to operation addition

Scanned by CamScanner

and for some $a, b \in Z \Rightarrow a+b \in Z$ (Closure)

$\therefore f(a+b) = 2(a+b)$ (∵ by defn. ①)

$= 2a + 2b$

$= f(a) + f(b)$ (∵ By defn ①)

$\Rightarrow$ f is Homomorphism.

Now since $f : (Z,+) \to (E,+)$ is Bijective and homomorphism

$\Rightarrow$ f is isomorphism

i.e. set Z is isomorphich with E

$\Rightarrow Z \cong E$

Ex: let $f : (R,+) \longrightarrow (R_+, \times)$ be a function define as $f(x) = e^x \quad \forall x \in R$

where R is Group of Real numbers & R_+ is Group of positive real numbers. under the operation addition and multiplication respectively.

Then show that R & R_+ are isomorphic.

Soln: let $(R,+)$ is group of real number w.r.to to operation addition.

& $(R_+, \times)$ is a group of positive real number w.r.to operation multiplication

and a function $f : (R,+) \longrightarrow (R_+, \times)$

is defined as $\boxed{f(x) = e^x}$ —① $\forall x \in R$

Scanned by CamScanner

To show f is one-one :

Let $x_1, x_2 \in R$ are any real numbers (-ve or +ve)

Such that $f(x_1)$ & $f(x_2) \in R_+$ (any positive real number)

where $f(x_1) = e^{x_1}$ & $f(x_2) = e^{x_2}$

Now consider that

$$x_1 \neq x_2 \Rightarrow e^{x_1} \neq e^{x_2}$$

$$\Rightarrow f(x_1) \neq f(x_2)$$

$\Rightarrow f$ is one-one

To show f is on-to

Since for $y \in R_+$ (any positive real number)

$\log y$ is any real number.

i.e. $\forall\ y \in R_+$ $\exists$ some $x \in R$ s.t.

$$x = \log y$$

$$\Rightarrow e^x = y$$

$$\Rightarrow y = f(x)$$

i.e. $\forall\ y \in R_+$ $\exists$ some $x \in R$ such that $y = f(x)$

$\therefore f$ is onto

To show f is Homomorphism

Let $(R, +)$ & $(R_+, \times)$ are groups under operations addition & multiplication R respectively.

Now, for $x, y \in R \Rightarrow R(x+y) \in R$ $\because$ (closure property w.r.to +)

Now By defn (1)

~~$\forall (x+y) \in R \Rightarrow f(x) = e^{(x+y)}$~~

$\forall\ (x+y) \in R \Rightarrow f(x+y) = e^{x+y}$

Scanned by CamScanner

$\Rightarrow f(x+y) = e^x \times e^y$

$= f(x) \times f(y)$

$\Rightarrow f(x+y) = f(x) \times f(y) \quad \forall x, y \in R$

$\therefore$ f is Homomorphism

Since f is Bijective (one-one - and onto) and f is also homomorphism.

$\Rightarrow$ a function $f: (R, +) \longrightarrow (R_+, \times)$ is isomorphism

i.e. group $(R, +)$ is isomorphic with $(R_+, \times)$

$\Rightarrow R \cong R_+$

Ex: let $(G, \cdot)$ be a group w.r.to multiplication and let a be the any fixed element of G prove that a function $f: G \longrightarrow G$. defined as $\forall x \in G$, $f(x) = axa^{-1}$ is isomorphism.

Soln: let $(G, \cdot)$ is a group w.r.to. multiplication and $f: G \longrightarrow G$ is a function from group $(G, \cdot)$ to $(G, \cdot)$. define as

$$\boxed{f(x) = a \cdot x \cdot a^{-1}} \quad \forall x \in G \ \& \ a \in G \quad \text{--- (1)}$$

where a is fixed element of G.

Scanned by CamScanner

To show f is one-one

Since $(G, \cdot)$ is group w.r.t multiplication operation.

$\therefore$ For every $x \in G$ & for any fixed element $a \in G$

$\Rightarrow x \cdot a \in G$ [closure property]

also since $a \in G$ & G is group.

its inverse is exist

let us consider a^{-1} is inverse of a and vice versa.

$\Rightarrow a^{-1} \cdot a = e = a \cdot a^{-1}$, where $e \in G$ is identity

Now since $a \in G$, $a^{-1} \in G$

$\therefore$ For any x & $y \in G$, $axa^{-1} \in G$ & $ayb^{-1} \in G$

Now. For $x \neq y \Rightarrow axa^{-1} \neq aya^{-1}$.

$\Rightarrow f(x) \neq f(y)$ (By defn ①)

$\Rightarrow f$ is one-one.

To show f is on-to :

Since $f: G \longrightarrow G$.

Domain of f and codomain of f are same set

$\therefore$ every element in Domain set G is also element of codomain set G

Now, for $x \in G$, $a \in G$, we know that

$a \cdot x \cdot a^{-1} \in G$ [closure]

let $y \in G$ is one of the element in codomain G such that $y = a \cdot x \cdot a^{-1} = f(x)$

$\therefore$ For every $y \in G$, $\exists$ some $x \in G$ such that $\boxed{y = f(x)} \Rightarrow f$ is onto

Scanned by CamScanner

To show: f is homomorphism.

let $\forall\ x, y \in G \Rightarrow x \cdot y \in G$ (closure)

Now,

$$f(x \cdot y) = a \cdot (xy) \cdot a^{-1} \quad (\because \text{By defn } ①)$$

$$= (a \cdot x) \cdot (y \cdot a^{-1}) \quad \because [\text{associative}]$$

$$= (a \cdot x) \cdot e \cdot (y \cdot a^{-1}) \quad [\text{identity } e \in G]$$

$$= (a \cdot x) \cdot (a^{-1} \cdot a)\ (y \cdot a^{-1}) \quad \because [a \cdot a^{-1} = e\ \forall\ a \in G]$$

$$= (a \cdot x \cdot a^{-1}) \cdot (a \cdot y \cdot a^{-1}) \quad [\text{Associative}]$$

$$\Rightarrow f(x \cdot y) = f(x) \cdot f(y) \quad (\text{By defn. } ①)$$

$\therefore$ f is homomorphism

Now, since f is Bijective (one-one & onto) and f is homomorphism.

$\Rightarrow$ $f : G \longrightarrow G$ is ~~iso~~ morphism

ie $G \cong G$.

Scanned by CamScanner